بسم الله الرحمن الرحيم

Data Mining (DM)

Fall 2025

Lecture 6a: Know your data

Rafia Asad Khan, University of Management and Technology, Lahore

Data Types, Data Statistics, Data Visualization

GETTING TO KNOW YOUR DATA

Introduction and Motivation

- What are the types of *attributes* or fields that make up your data?
- What kind of values does each attribute have?
- Which attributes are discrete, and which are continuous-valued?
- What do the data *look like*? How are the values distributed?
- Are there ways we can visualize the data to get a better sense of it all?
- Can we spot any outliers?
- Can we measure the similarity of some data objects with respect to others?

Chapter 2: Getting to Know Your Data

- Data Objects and Attribute Types
- Basic Statistical Descriptions of Data
- Data Visualization
- Measuring Data Similarity and Dissimilarity
- Summary

Data Objects

- Data Object
  - Represents an <u>entity</u>

- Sales database
  - Customers
  - Store items
  - Sales

Data Objects

- Data Object
  - Represents an <u>entity</u>

- Medical database
  - Patients
  - Doctors
  - …

Data Objects

- Data Object
  - Represents an entity
- University database
  - Students
  - Professors
  - Courses

Data Objects

- Data objects can also be referred to as
  - *Samples*
  - *Examples*
  - *Instances*
  - *Data points*
  - *Objects*
  - *Tuples*

Data Objects

- Data Object
  - Represents an <u>entity</u>

- **Attributes**
  - Used to <u>describe</u> Data objects

Attribute

- An attribute is a data field, *representing* a **characteristic** or **feature** of a data object

- A.k.a.
  - *Dimension*
  - *Feature*
  - *Variable*

Attribute

- An attribute is a data field, *representing* a **characteristic** or **feature** of a data object
- A.k.a.
  - *Dimension* *(Data Warehouse)*
  - *Feature* *(Machine Learning)*
  - *Variable* *(Statistics)*

Attribute

- Customer (object)
  - *Customer ID*
  - *Name*
  - *Address*

- *Student (object)*
  - *???*
  - *???*
  - …

Observations

- Observed **values** for a given attribute are known as *observations*

Student Object

- **Attributes**
  - Name
  - ID
  - CGPA
- **Observations**
  - Mushtaq
  - F2012065219
  - 3.84

  - Hamna
  - S2011599031
  - 3.77

Attribute Vector (or Feature Vector)

- A ***set of attributes*** used to describe a given object

- Customer Object
  - Customer ID, Name, Address

- Student Object
  - Name, ID, CGPA

Types of Attributes

- Nominal
- Binary
- Ordinal
- Numeric

Nominal Attributes

- Relating to Names
- Values of a nominal attribute are
  - **Symbols**, or
  - ***names of things***
  - e.g. category, code, or state
- A.k.a. **Categorical Attributes**

Nominal Attributes

- Values of a nominal attribute are
  - **Symbols,** or
  - ***names of things***
  - e.g. category, code, or state

- *Category*
  - Undergraduate, Graduate
- *Code*
  - 065, 266, 105
- *State*
  - Present, Absent

Example 2.1 **Nominal attributes.** Suppose that *hair_color* and *marital_status* are two attributes describing *person* objects. In our application, possible values for *hair_color* are *black*, *brown*, *blond*, *red*, *auburn*, *gray*, and *white*. The attribute *marital_status* can take on the values *single*, *married*, *divorced*, and *widowed*. Both *hair_color* and *marital_status* are nominal attributes. Another example of a nominal attribute is *occupation*, with the values *teacher*, *dentist*, *programmer*, *farmer*, and so on. ■

Binary Attributes

- A *nominal attribute* with **only two** categories or states: 0 or 1

- 0 typically means that the attribute is absent, and 1 means that it is present

- A.k.a.
  - *Boolean Attributes*
    - if the two states correspond to ***true*** *and* ***false***

Example 2.2 **Binary attributes.** Given the attribute *smoker* describing a *patient* object, 1 indicates that the patient smokes, while 0 indicates that the patient does not. Similarly, suppose the patient undergoes a medical test that has two possible outcomes. The attribute *medical_test* is binary, where a value of 1 means the result of the test for the patient is positive, while 0 means the result is negative. ■

Binary Attributes

- **Symmetric**
  - If both of its states are **equally valuable** and carry the same weight
    - That is, there is no preference on which outcome should be coded as 0 or 1
  - One such example could be the attribute *gender* having the states **male** and **female**

Binary Attributes

- **Asymmetric**
  - If both of its states are *NOT* **equally valuable**
  - Such as the *positive* and *negative* outcomes of a medical test for HIV

Binary Attributes

- **Asymmetric**
  - By convention, we code the **most important** outcome, by 1
    - Which is usually the ***rarest*** one (e.g., *HIV positive*)
  - And the other by 0
    - (e.g., *HIV negative*)

Ordinal Attributes

- An attribute with possible values that have a meaningful **order** or **ranking** among them
- But the magnitude between successive values is not known
- For example, A Pizza ☺
  - *small, medium, large*

Ordinal Attributes

- A Pizza ☺
  - *small, medium, large*
- *Drink Size :*
  - *small, medium, and large*
- *Grade :*
  - *A+, A, A-, B+,* and so on

Ordinal Attributes

- Ordinal attributes are often used in surveys for Ratings
  - *0: very dissatisfied, 1: somewhat dissatisfied, 2: neutral, 3: satisfied,* and *4: very satisfied.*

Qualitative (*Aspect of*) Data

- Nominal
- Binary
- Ordinal

Numeric Attributes

- A numeric attribute is **quantitative**
  - i.e. It is a *measurable* quantity
  - Represented in
    - ***integer*** or ***real*** values
- Numeric attributes can be
  - Interval-scaled
  - Ratio-scaled

Numeric Attributes

- Interval-scaled
  - Measured on a scale of **equal-size** units
  - The values can be *positive*, **0**, or *negative*

Numeric Attributes

- Interval-scaled
  - Not only allow to ***compare,*** but also to quantify the ***difference*** between values
  - However, we can ***not*** speak of a value as being a multiple (or ***ratio***) of another value
    - For example, we ***cannot*** say that **10° C is *twice* as warm as 5° C**

Example 2.4 **Interval-scaled attributes.** A *temperature* attribute is interval-scaled. Suppose that we have the outdoor *temperature* value for a number of different days, where each day is an object. By ordering the values, we obtain a ranking of the objects with respect to *temperature*. In addition, we can quantify the difference between values. For example, a temperature of 20°C is five degrees higher than a temperature of 15°C. Calendar dates are another example. For instance, the years 2002 and 2010 are eight years apart. ■

Numeric Attributes

- Ratio-scaled
  - Inherent **zero-point**
  - So, we can speak of a value as being a **multiple** (or **ratio**) of another value
  - These too, allow to ***compare,*** as well as quantify the ***difference*** between values

Example 2.5 **Ratio-scaled attributes.** Unlike temperatures in Celsius and Fahrenheit, the Kelvin (K) temperature scale has what is considered a true zero-point ($0^\circ K = -273.15^\circ C$): It is the point at which the particles that comprise matter have zero kinetic energy. Other examples of ratio-scaled attributes include *count* attributes such as *years_of_experience* (e.g., the objects are employees) and *number_of_words* (e.g., the objects are documents). Additional examples include attributes to measure weight, height, latitude and longitude coordinates (e.g., when clustering houses), and monetary quantities (e.g., you are 100 times richer with $100 than with $1). ■

Discrete vs. Continuous Attributes

- **Discrete Attribute**
  - Has only a finite or countably infinite set of values
    - E.g., zip codes, profession, or the set of words in a collection of documents
  - Sometimes, represented as integer variables
  - Note: Binary attributes are a special case of discrete attributes
- **Continuous Attribute**
  - Has real numbers as attribute values
    - E.g., temperature, height, or weight
  - Practically, real values can only be measured and represented using a finite number of digits
  - Continuous attributes are typically represented as floating-point variables

Data set

- A **data set** is a collection of numbers or values that relate to a particular subject or task.
  - The test scores of each student in a particular class
  - The transactions at a store
- Data set for a student's result prediction?

Student's Course Grade Data set

- Example Data set for a student's Course Grade prediction
- Attributes
    - Course-code
    - Course-title
    - Course-type
    - Course-stream
    - Semester-number
    - Semester
    - Year
    - Number of registered courses in semester
    - PhD teacher
    - Male teacher
    - Grade

Standard Public Data sets

- Standard Public Data sets available online
  - Kaggle
  - UCI Machine Learning Repository
  - Open Data Pakistan
  - Google Trends
  - …